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Launch a Kubernetes Cluster



Ahmed Tetteh

The screenshot displays the Amazon Elastic Kubernetes Service (EKS) console interface. At the top, a green notification bar states "Access entry was successfully created." Below this, the cluster name "nextwork-eks-cluster" is shown with buttons for "Delete cluster", "Upgrade version", and "Monitor cluster". A warning banner indicates that standard support for Kubernetes version 1.33 ends on July 29, 2026, with an "Upgrade" button.

The "Cluster info" section provides a summary of the cluster's status and health:

- Status:** Active
- Kubernetes version:** 1.33
- Support period:** Standard support until July 29, 2026
- Provider:** EKS
- Cluster health:** 0 issues
- Upgrade insights:** 5 updates available
- Node health issues:** 0 issues
- Capability issues:** 0 issues

The "Nodes" section shows a table of three active nodes, all managed by the "nextwork-nodegroup".

Node name	Instance type	Compute	Managed by	Created	Status
ip-192-168-18-234-ap-northeast-1.compute.internal	t3.micro	Node group	nextwork-nodegroup	an hour ago	Ready
ip-192-168-41-125-ap-northeast-1.compute.internal	t3.micro	Node group	nextwork-nodegroup	an hour ago	Ready
ip-192-168-74-35-ap-northeast-1.compute.internal	t3.micro	Node group	nextwork-nodegroup	an hour ago	Ready

The "Node groups" section shows a single node group named "nextwork-nodegroup" with a desired size of 3, using AMI release version 1.33.5-20251217 and the launch template "eksctl-nextwork-eks-cluster-nodegroup-nextwork-nodegroup (1)".

Group name	Desired size	AMI release version	Launch template	Status
nextwork-nodegroup	3	1.33.5-20251217	eksctl-nextwork-eks-cluster-nodegroup-nextwork-nodegroup (1)	Active



Ahmed Tetteh
NextWork Student

nextwork.org

Introducing Today's Project!

In this project, I will deploy my very own Kubernetes cluster using Amazon EKS , and monitor the cluster creation via CloudFormation. I will also be using some be using some interesting tools like eksctl and IAM access entries.

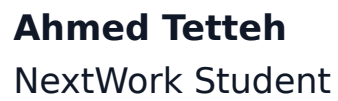


What is Kubernetes?

Kubernetes is a container orchestrator tool that manages containers in terms of scaling to meet workloads, restarting them when they stop and deploying them across servers. It makes it easy to manage large number of containers across several devices automatically, by setting up some few instructions, and kubernetes handles it from there. Companies and developers use Kubernetes to to manage their applications running on containers, as it makes their application management much more steady and easier to scale when traffic increases.

I used eksctl to to interact with Amazon EKS and create a cluster. The create cluster command I ran defined the cluster name I want to create, the node group, the region I want the cluster to be created in, the type of node that should run in my cluster, and even the version of Kubernetes I will be using.

I initially ran into two errors while using eksctl. The first one was because. I didn't have the tool "eksctl" installed on my instance(virtual server). The second one was because my EC2 instance didn't have the permissions or credentials to create an EKS cluster.



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Amazon Linux 2023

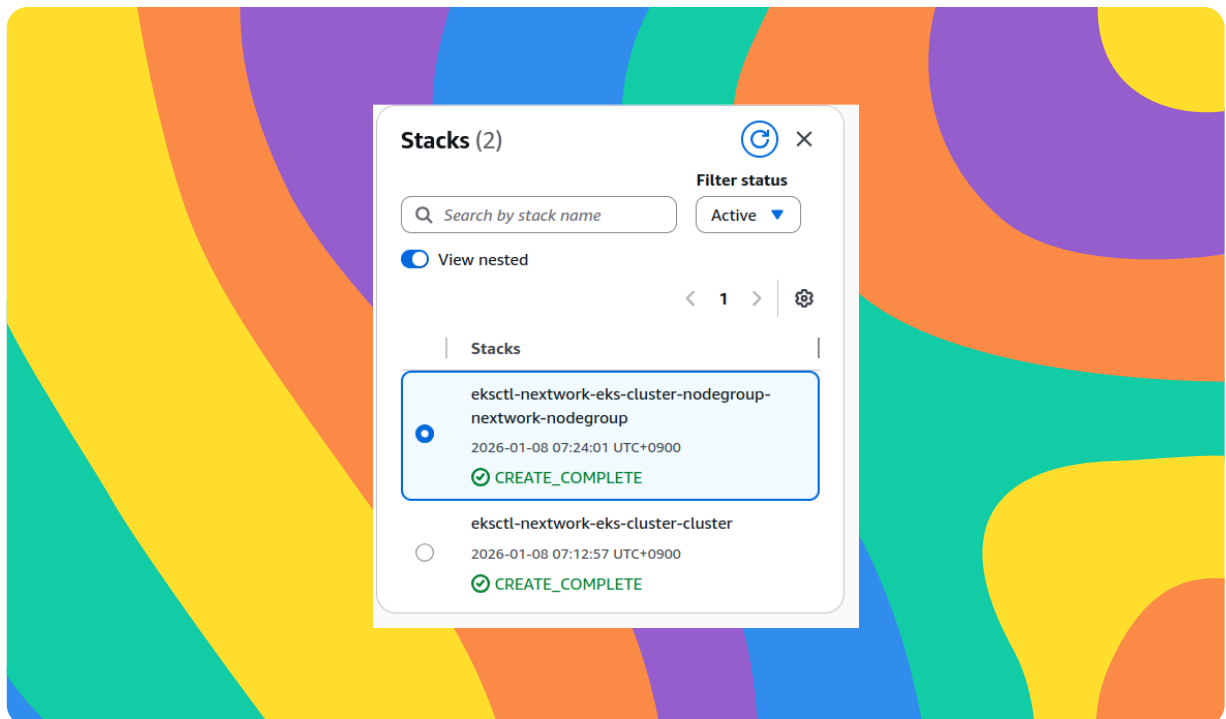
https://aws.amazon.com/linux/amazon-linux-2023


Last login: Wed Jan 7 21:27:48 2026 from 3.112.23.5
[ec2-user@ip-172-31-46-145 ~]$ eksctl create cluster --name network-eks-cluster --nodegroup-name network-nodegroup --node-type t3.micro --nodes 3 --nodes-min 1 --nodes-max 3 --version 1.33 --region ap-northeast-1
--bash: eksctl: command not found
[ec2-user@ip-172-31-46-145 ~]$
```


eksctl and CloudFormation

CloudFormation helped create my EKS cluster because it takes instructions from eksctl to automatically provision all the resources I need for my eks cluster. It created VPC resources because my the pods and node in the eks cluster needs some specific networking resources to communicate with other and pass traffic.

There was also a second CloudFormation stack for my nodegroup, i.e. the servers actually running my containers or pods. The difference between a cluster and node group is that my cluster is the entire environment managed by kubernetes (the control plane and node agents), while the node group consists of a collection of nodes running my containers or pods.





The EKS console

I had to create an IAM access entry in order to have the permissions to access my nodes, since Kubernetes comes with its own permission setup. An access entry allows an IAM user or role to access the Kubernetes cluster by mapping the IAM user to Kubernetes's own RBAC (Role Based Access Control). I set it up by selecting my current IAM user and adding this policy: "AmazonEKSClusterAdminPolicy". This allows my IAM user full access to the EKS cluster, including the nodes running the pods or containers.

It took me about 1 hour, including setup and background studies on clusters to create my cluster. Since I'll create this cluster again in the next project of this series, maybe this process could be sped up if I quickly move on to the next part.



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The "Cluster info" section provides a summary of the cluster's status and configuration:

- Status:** Active
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- Upgrade insights:** 5 updates
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- Capability issues:** 0 issues

The "Nodes" section shows a table of 3 nodes, all in a "Ready" state:

Node name	Instance type	Compute	Managed by	Created	Status
ip-192-168-18-234.ap-northeast-1.compute.internal	t3.micro	Node group	nextwork-nodegroup	an hour ago	Ready
ip-192-168-41-125.ap-northeast-1.compute.internal	t3.micro	Node group	nextwork-nodegroup	an hour ago	Ready
ip-192-168-74-35.ap-northeast-1.compute.internal	t3.micro	Node group	nextwork-nodegroup	an hour ago	Ready

The "Node groups" section shows a table with 1 node group, which is "Active":

Group name	Desired size	AMI release version	Launch template	Status
nextwork-nodegroup	3	1.33.5-20251217	eksctl-nextwork-eks-cluster-nodegroup-nextwork-nodegroup (1)	Active



EXTRA: Deleting nodes

Did you know you can find your EKS cluster's nodes in Amazon EC2? This is because the nodes themselves are EC2 instances(virtual computing resources) created by CloudFormation, following instructions from the eksctl tool. These instances or nodes are then managed by Amazon EKS for running pods and containers.

Desired size means the number of nodes I want running in my node group. Minimum and maximum sizes are helpful for scaling down or up when traffic or the workload on the nodes is low or high. Kubernetes then provisions these nodes based on those configurations to keep the application always available.

When I deleted my EC2 instances, they were also deleted from my nodegroup, which Kubernetes immediately noticed this change and provisioned new instances. This is because of the configurations I made for the number of nodes in my nodegroup.



Instances (3/7) Info Last updated 1 minute ago [Connect](#) [Instance state](#) [Actions](#) [Launch Instances](#)

[All states](#)

☐	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone
☐	nextwork-eks-ints	i-0cd248774b8d8e783	Running	t3.micro	3/3 checks passed	View alarms	ap-northeast-1a
☐	nextwork-eks-cluster-nextwork-nodegro...	i-0c59bc1ea26fb43da	Terminated	t3.micro	-	View alarms	ap-northeast-1a
☒	nextwork-eks-cluster-nextwork-nodegro...	i-04eee7646f135fa2d	Running	t3.micro	3/3 checks passed	View alarms	ap-northeast-1a
☐	nextwork-eks-cluster-nextwork-nodegro...	i-022a3b7cf2c1623d3	Terminated	t3.micro	-	View alarms	ap-northeast-1d
☒	nextwork-eks-cluster-nextwork-nodegro...	i-06a264c80158dc846	Running	t3.micro	3/3 checks passed	View alarms	ap-northeast-1d
☐	nextwork-eks-cluster-nextwork-nodegro...	i-0b11658ec7e27e5eb	Terminated	t3.micro	-	View alarms	ap-northeast-1c
☒	nextwork-eks-cluster-nextwork-nodegro...	i-0c7a4143e16705b39	Running	t3.micro	3/3 checks passed	View alarms	ap-northeast-1c



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